A fixed testing family on the inverse-Alice chart

23 October 2026

Theorem 1. *Put $\delta = 2^{-12}$. Suppose $0 < b < 1$, $Q = 1 - b$, $0 < z \le 1/64$, $q = \sqrt{z}$, and the source coefficients satisfy*

$$m, D, \beta \ge 0, \qquad C \ge C_* = \frac{3z}{8}, \quad A \ge A_* = \frac{3}{8bz}, \quad \alpha \ge \alpha_* = \frac{3}{8Qz}.$$

For $0 \le p \le 1$, $0 \le w \le \delta$, and every real state, the complete inverse-Alice C1 parent below has an SOS in one fixed source-independent family of 51 labelled testing functions. The construction also uses 27 polynomial weighted positive semidefinite source blocks of total order 66, maximum order six. There is no restriction on H or the sign of λ. All source faces $D = 0, \beta = 0$ and all testing/state zeros are included. The same identities hold over independent real closed source and testing fields; all Gram factorizations below take place in the source field.

Write $D_p = 1 + p^2$, $D_w = 1 + w^2$. The full parent, with $d = 1$, is

$$\begin{aligned}\widetilde{F} &= 4m(pW - pwX + Y - wZ)^2 + 4CD_w(W - pY)^2 + 4AD_pD_wY^2 \\ &\quad + (4D + 2)(wW + X - pwY - pZ)^2 + (4\alpha + 2)D_p(Y - wZ)^2 \\ &\quad + (4\beta + 2)D_w(pX + Z)^2 - 2D_pD_wZ^2.\end{aligned}$$

It equals $p^2F_w(1/p, w)$ when $p \ne 0$ and defines the literal $p = 0$ angle. Introduce five separated state roots and one source change of coordinates:

$$E = qW, \quad F = Y/q, \quad G = wZ/q, \quad R = X + wW, \quad S = pX, \qquad J = F - bG.$$

Their sizes need not be bounded. The complete signed core is $2K$, where direct expansion gives

$$K = R^2 + D_wS^2 + z(G - F)^2 + p^2D_wzF^2 - 2pE(G + w^2F) + 2qwS(G - F).$$

Select just the three caps C_*, A_*, α_*. Put $t_J = 3/(2bQ) \ge 6$. Expanding these caps and $2K$ gives

$$\widetilde{F} = B + \mathcal{R} + \mathcal{P} + \mathcal{X},$$

where $\mathcal{X}$ is the six original event terms with weights $4m, 4D, 4\beta, 4(C - C_*), 4(A - A_*), 4(\alpha - \alpha_*)$, including their complete D_p, D_w factors. Furthermore

$$\begin{aligned}B &= 2R^2 + 2S^2 + J^2 + \tfrac{3}{2}E^2 + \tfrac{3}{2}(1 + p^2)G^2 - 4pEG, \\ \mathcal{R} &= -(3pz + 4pw^2)E(J + bG) - 4qwS(J - QG), \\ \mathcal{P} &= (t_J - 1)J^2 + t_J(pJ)^2 + \tfrac{3}{2}z^2(pF)^2 + \tfrac{3}{2}\{w(E - pzF)\}^2 \\ &\quad + \frac{3}{2b}\{(wF)^2 + (pwF)^2\} + 2(wS)^2 + 2z(G - F)^2 + 2zD_w(pF)^2.\end{aligned}$$

Every coefficient in $\mathcal{P}$ is positive and source-only. Thus these are ten retained squares; expanding $\mathcal{X}$ gives twelve further squares.

A fixed reserve and three error channels. On the ordered roots $g = (R, S, J, E, G)$,

$$B - \tfrac{1}{9} g^{\mathsf{T}} g = \frac{17}{9}(R^2 + S^2) + \frac{8}{9} J^2 + \frac{25}{18}(E - 36pG/25)^2 + \frac{2}{225} G^2 + \frac{69}{50}(1 - p^2) G^2.$$

This is six explicit squares with only the fixed testing radical $\sqrt{1 - p^2}$. Set

$$y_1 = p, \qquad y_2 = p(w/\delta)^2, \qquad y_3 = w/\delta.$$

All lie in $[0, 1]$. Write $\mathcal{R} = \sum_{j=1}^{3} y_j g^T A_j g$. The only upper-triangle entries of these symmetric source matrices are

	entry	value	r_j
A_1	$(E, J), (E, G)$	$-3z/2, -3bz/2$	$3z$
A_2	$(E, J), (E, G)$	$-2\delta^2, -2b\delta^2$	$4\delta^2$
A_3	$(S, J), (S, G)$	$-2q\delta, 2qQ\delta$	$4q\delta.$

Their absolute row sums are at most r_j, so $r_j I \pm A_j \succeq 0$ in the source field. Rational arithmetic gives

$$\sum_j r_j \le 3/64 + 4\delta^2 + \delta/2 < 1/18.$$

Hence $H_0 = (1/9 - \sum r_j) I \succeq I/18$ and $H_j^{\pm} = (r_j I \pm A_j)/2 \succeq 0$. Exactly,

$$\tfrac{1}{9} I + \sum_j y_j A_j = H_0 + \sum_j \{(1 + y_j) H_j^{+} + (1 - y_j) H_j^{-}\}.$$

Factor these seven matrices over the source real closed field. Only the six fixed testing radicals $\sqrt{1 \pm y_j}$ are added; no mixed radical, adaptive constant, or source/testing comparison occurs.

The fixed family. Let $\phi_6 = (X, Y, W, wW, wZ, pX)$. The roots g are source-linear combinations of ϕ_6. The 42 functions ϕ_6 times $1, \sqrt{1 \pm y_1}, \sqrt{1 \pm y_2}, \sqrt{1 \pm y_3}$ span this seven-block contribution. Add

$$pwZ,\ pY,\ pwY,\ wY,\ pwX,\ pW,\ pZ,\ Z, \quad wZ\sqrt{1 - p^2}.$$

These nine functions contain all six baseline roots and all twenty-two retained roots. This proves the 51-function count. The polynomial weighted form keeps the seven main vectors on ϕ_6, six baseline vectors, eight grouped selected-positive vectors, and six whole event vectors. Their orders sum to $42 + 8 + 10 + 6 = 66$.

Native source admission. Here the opening numerator is denoted π to distinguish it from the testing coordinate p. For the native C1 formulas put

$$r = bz, \quad \theta = 1 - h, \quad e = xh/\sigma, \quad E_s = 1 - e > 0, \quad K = 1 + x, \quad R_s = 1 + xr,$$
$$M = R_s - xz\theta > 0, \quad J_s = 1 + xh - \sigma(1 - r) > 0,$$
$$P = \sigma + K + \sigma z(1 + bx + 2xh) + \sigma b x^2 h z^2 + x^2 h \theta z,$$
$$\pi = \frac{z(1 + xh)E_s + (1 - r)(1 + xhz)}{2P}, \quad l = \frac{(1 + \sigma r)\{eR_s(1 + \sigma z) + ME_s\}}{2P},$$
$$C = \frac{Kz(1 + \sigma r)E_s}{2(1 - r)P}, \quad A = \frac{\pi}{bz}, \quad \alpha = \frac{l}{Qz},$$
$$D = \frac{K(1 + \sigma z)(1 + \sigma r) - J_s P}{2J_s P}, \quad \beta = \frac{K(1 + \sigma r)(1 + xhz) - \theta P}{2\theta P}.$$

Take $0 < x, \sigma \le \epsilon = 2^{-30}$, $0 < h, b < 1$, and the native nonnegative D, β. With $H_0 = 1 + zQ$, $B_s = P + xz(J_s - \sigma)$, the source identities give

$$eB_s = xH_0 + \sigma\mathcal{H} - 2PJ_sD/\sigma,$$

$$\mathcal{H} = R_sH_0 + z\{(1-r)xh(2+xr) - e[2 + x(1+h+r\theta)] + Ke^2\}.$$

Initially $0 < e < 1$, $1 \le P \le 1 + 5\epsilon$, $B_s \ge 1$, $M \ge 1 - \epsilon$, and $\mathcal{H} \le 4$. Thus $e \le 6\epsilon$. Consequently

$$C/z,\ \pi\ \ge \frac{1 - 6\epsilon}{2(1 + 5\epsilon)} > 3/8, \qquad l \ge \frac{1 - \epsilon}{2(1 + 5\epsilon)} > 3/8.$$

For π use its numerator $\ge H_0 - ze$; for l use its numerator $\ge e + M(1 - e) \ge 1 - \epsilon$. These prove the three caps without any bound on $H = h/\sigma$, b^{-1}, or Q^{-1}.

Restoration and literal boundaries. The two exact angle sign gauges give signed p, w through their fixed absolute-angle/state-sign map. On $p \neq 0$ the physical slack for unit states is $\widetilde{F}/(D_pD_w)$. Multiply every testing function by the pure-testing κ_t, where $\kappa_t^2 D_pD_w = 1$; this remains well-defined at $p = 0$. All final vectors are polynomial in state and testing coordinates before whole-block gate radicals. Thus $p = 0, w = 0$ and every state-coordinate zero are literal specializations. Gate signs multiply whole blocks and are harmless. Real and imaginary state parts give complex states. This proves a local fixed-family admission theorem on an additional testing chart, not a global quantum-set lift.